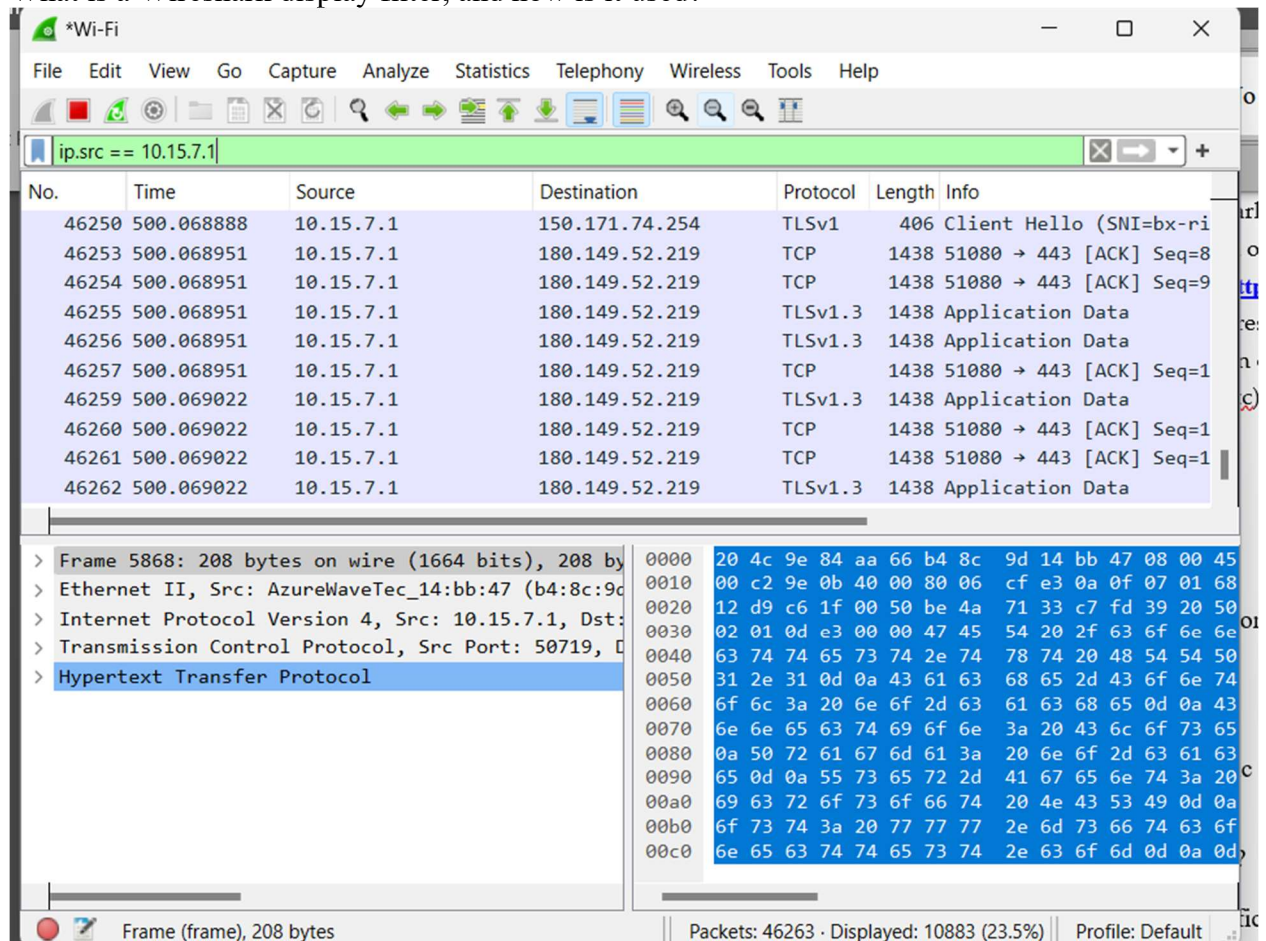


CN Assignment-2

Taking Wireshark for a Test Run: Tutorial on basics of Wireshark packet sniffer tools is available for download here. You need to first install Linux version of Wireshark software in your computer in the lab. Wireshark is available for download at <http://www.wireshark.org/>. In this assignment, you will be tested for your familiarity in using Wireshark for packet capture and analysis. Start packet capture in wireshark application and then open your web browser and type in an URL of website of your choice (www.google.com, etc). Answer the following questions, based on your experimentation:

Replace IP with your own source IP. Also attach screenshots.

- 1) What is a Wireshark display filter, and how is it used?



CN Assignment-2

The image shows a Wireshark packet capture window titled '*Wi-Fi'. The filter bar at the top contains the expression 'ip.dst == 10.15.7.1'. The packet list pane displays several packets, with packet 50983 selected. The packet details pane shows the structure of the selected packet: Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol. The packet bytes pane shows the raw data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
50958	531.587931	151.101.38.172	10.15.7.1	TCP	1510	[TCP Previous segment not c
50960	531.636531	54.192.142.4	10.15.7.1	TCP	68	[TCP Keep-Alive ACK] 443 →
50961	531.646228	151.101.38.172	10.15.7.1	TCP	1510	80 → 51045 [ACK] Seq=305593
50962	531.651545	151.101.38.172	10.15.7.1	TCP	1438	[TCP Window Full] 80 → 5104
50968	531.793286	151.101.38.172	10.15.7.1	TCP	1510	[TCP Out-Of-Order] 80 → 510
50970	531.907813	151.101.38.172	10.15.7.1	TCP	282	80 → 51045 [PSH, ACK] Seq=3
50971	531.907813	151.101.38.172	10.15.7.1	TCP	1222	[TCP Window Full] [TCP Prev
50977	532.020176	151.101.38.172	10.15.7.1	TCP	1510	[TCP Out-Of-Order] 80 → 510
50979	532.076014	142.250.194.78	10.15.7.1	UDP	73	443 → 50386 Len=31
50983	532.144885	151.101.38.172	10.15.7.1	TCP	1510	80 → 51044 [ACK] Seq=231177

Frame 5863: 1466 bytes on wire (11728 bits), 1466 bytes captured (11728 bits) on interface 0

Ethernet II, Src: Cisco_84:aa:66 (20:4c:9e:84:aa:66), Dst: 10.15.7.1

Internet Protocol Version 4, Src: 142.250.194.42, Dst: 10.15.7.1

Transmission Control Protocol, Src Port: 443, Dst Port: 51044

0000 b4 8c 9d 14 bb 47 20 4c 9e 84 aa 66 08 00 00 45
 0010 05 ac f5 0a 00 00 38 06 26 0d 8e fa c2 2a 0e
 0020 07 01 01 bb c6 1c 68 42 c5 ce e1 c7 f4 cc 56
 0030 01 17 28 ac 00 00 e0 a4 61 bf f2 f9 27 69 91
 0040 4e 54 f8 70 e2 8a 8c 87 bf d5 5f 3f a5 d4 c5
 0050 fb f4 c0 f6 0f 7b 58 f7 53 db 99 a6 9b cb a2
 0060 65 d5 fe c0 a0 2d 45 4b bd 74 32 a4 f4 d8 ba
 0070 cd 3b 6c 54 50 20 d0 9c fd fa 47 22 e3 06 41
 0080 30 c3 7b 6c e3 47 37 92 94 06 16 f1 88 17 11
 0090 cc 6e 04 a9 59 1c 83 60 34 f8 7f 2d 7f 51 a8
 00a0 35 2c e5 50 07 80 4f 48 e6 41 cf 76 15 be 81
 00b0 5b 31 a1 e3 bf 40 8d a5 fe 6b bc c1 66 50 c3
 00c0 65 7b 2a 9c cb 34 36 e0 e0 a2 80 4a 0a 66 31
 00d0 30 60 e0 01 54 53 32 21 44 3b f8 1c f9 44 0e

Destination Address: IPv4 address | Packets: 50983 · Displayed: 12514 (24.5%) | Profile: Default

- 2) How would you write a display filter to show only packets coming from a specific IP address, such as 192.168.1.1?

Ip.src == <ip address>

CN Assignment-2

The image shows a Wireshark network traffic capture window. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. Below the menu is a toolbar with various icons. A filter bar at the top displays the filter `ip.src == 172.16.1.6`. The main packet list table shows the following data:

No.	Time	Source	Destination	Protocol	Length	Info
71905	709.074547	172.16.1.6	10.15.7.1	TCP	1514	[TCP Spurious Retransmission]
71912	709.074547	172.16.1.6	10.15.7.1	TCP	1514	[TCP Spurious Retransmission]
71913	709.074547	172.16.1.6	10.15.7.1	TCP	1514	80 → 51238 [ACK] Seq=200021
71916	709.074547	172.16.1.6	10.15.7.1	TCP	1514	80 → 51245 [ACK] Seq=94901 /
71917	709.074547	172.16.1.6	10.15.7.1	TCP	1514	80 → 51245 [ACK] Seq=96361 /
71920	709.074547	172.16.1.6	10.15.7.1	TCP	1514	80 → 51238 [ACK] Seq=201481
72205	710.247450	172.16.1.6	10.15.7.1	TCP	1514	80 → 51245 [ACK] Seq=97821 /
72206	710.247578	172.16.1.6	10.15.7.1	TCP	1514	80 → 51245 [ACK] Seq=99281 /
72208	710.248427	172.16.1.6	10.15.7.1	TCP	1514	80 → 51245 [ACK] Seq=100741
72209	710.248427	172.16.1.6	10.15.7.1	TCP	1514	[TCP Previous segment not c]

The detailed view of the selected packet (No. 72209) shows the following structure:

- > Frame 62333: 68 bytes on wire (544 bits), 68 bytes
- > Ethernet II, Src: Cisco_84:aa:66 (20:4c:9e:84:aa:66)
- > Internet Protocol Version 4, Src: 172.16.1.6, Dst:
- > Transmission Control Protocol, Src Port: 80, Dst Po

The packet bytes are displayed in hexadecimal and ASCII format:

```
0000 b4 8c 9d 14 bb 47 20 4c 9e 84 aa 66 08 00 45 06
0010 00 34 00 00 40 00 3e 06 7e 9e ac 10 01 06 0a 01
0020 07 01 00 50 c7 eb 44 d7 c0 9e cc f6 6b 23 80 17
0030 16 d0 94 3f 00 00 02 04 05 b4 01 01 04 02 01 03
0040 03 07 e7 44
```

The status bar at the bottom indicates: Frame (frame), 68 bytes | Packets: 72234 · Displayed: 843 (1.2%) | Profile: Default

3) Write a display filter to show only HTTP traffic.

CN Assignment-2

The image shows a Wireshark packet capture window. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for packet capture and analysis. The packet list table is displayed with the following columns: No., Time, Source, Destination, Protocol, Length, and Info.

No.	Time	Source	Destination	Protocol	Length	Info
75292	734.466605	49.44.179.41	10.15.7.1	HTTP	523	HTTP/1.1 301 Moved Permanent
75495	735.839949	172.16.1.6	10.15.7.1	HTTP	1514	[TCP Previous segment not ca
75535	736.104996	172.16.1.6	10.15.7.1	HTTP	1514	[TCP Previous segment not ca
75535	736.104996	172.16.1.6	10.15.7.1	HTTP	1514	[TCP Previous segment not ca
75594	736.810385	172.16.1.6	10.15.7.1	HTTP	1514	[TCP Previous segment not ca
75605	736.811058	10.15.7.1	172.16.1.6	HTTP	578	GET /media/mod_vt_nivo_slide
75620	736.865489	172.16.1.6	10.15.7.1	HTTP	582	HTTP/1.1 200 OK (GIF89a)
75659	737.196687	10.15.7.1	172.16.1.6	HTTP	543	GET /templates/newsite/image
76052	740.177108	172.16.1.6	10.15.7.1	HTTP	1141	HTTP/1.1 200 OK (PNG)
76112	740.418198	10.15.7.1	172.16.1.6	HTTP	542	GET /templates/newsite/image

The packet details pane shows the structure of a selected packet (Frame 53796):

- > Frame 53796: 317 bytes on wire (2536 bits), 317 bytes captured (2536 bits) on interface 0
- > Ethernet II, Src: Cisco_84:aa:66 (20:4c:9e:84:aa:66), Dst: 10:15:7:1 (08:00:2b:10:15:7:1)
- > Internet Protocol Version 4, Src: 49.44.102.184, Dst: 10.15.7.1
- > Transmission Control Protocol, Src Port: 80, Dst Port: 443
- > Hypertext Transfer Protocol

The packet bytes pane shows the raw data of the selected packet:

```
0000 b4 8c 9d 14 bb 47 20 4c 9e 84 aa 66 08 00 45
0010 01 2f 59 4b 40 00 31 06 46 8a 31 2c 66 b8 0a
0020 07 01 00 50 c7 ca 8d 7c 2c ab 66 2a e3 c6 50
0030 01 f5 be d5 00 00 48 54 54 50 2f 31 2e 31 20
0040 30 34 20 4e 6f 74 20 4d 6f 64 69 66 69 65 64
0050 0a 43 6f 6e 74 65 6e 74 2d 54 79 70 65 3a 20
0060 70 70 6c 69 63 61 74 69 6f 6e 2f 70 6b 69 78
0070 63 72 6c 0d 0a 4c 61 73 74 2d 4d 6f 64 69 66
0080 65 64 3a 20 4d 6f 6e 2c 20 31 32 20 46 65 62
0090 32 30 32 34 20 32 32 3a 30 37 3a 32 37 20 47
00a0 54 0d 0a 45 54 61 67 3a 20 22 36 35 63 61 39
00b0 39 66 2d 31 32 62 22 0d 0a 43 61 63 68 65 2d
00c0 6f 6e 74 72 6f 6c 3a 20 6d 61 78 2d 61 67 65
00d0 33 36 30 30 0d 0a 45 78 70 69 72 65 73 3a 20
```

- 4) How can you filter packets to show only those with a specific TCP port, such as port 443?

CN Assignment-2

Wireshark capture showing a filter `tcp.dstport == 443`. The packet list displays several TCP segments, including a retransmission and a duplicate ACK. The packet details pane shows the structure of a TCP segment for packet 1925.

No.	Time	Source	Destination	Protocol	Length	Info
1924...	1300.281901	10.15.7.1	180.149.59.145	TLSv1.2	96	Application Data
1924...	1300.281953	10.15.7.1	49.44.178.186	TLSv1.2	96	Application Data
1924...	1300.402558	10.15.7.1	3.218.189.67	TCP	54	51557 → 443 [ACK] Seq=4939 Ack=123030 Win=131328 Len=0
1924...	1300.402989	10.15.7.1	142.250.194.195	TCP	54	51952 → 443 [ACK] Seq=2601 Ack=6929 Win=130560 Len=0
1924...	1300.403530	10.15.7.1	142.250.194.195	TCP	54	51952 → 443 [ACK] Seq=2601 Ack=11165 Win=131072 Len=0
1924...	1300.404869	10.15.7.1	3.210.94.245	TLSv1.2	310	Application Data
1925...	1300.453812	10.15.7.1	3.218.189.67	TCP	54	51557 → 443 [ACK] Seq=4939 Ack=124490 Win=131328 Len=0
1925...	1300.469384	10.15.7.1	142.250.194.195	TCP	85	[TCP Retransmission] 51952 → 443 [PSH, ACK] Seq=2570 Ack=11165 Win=13...
1925...	1300.485193	10.15.7.1	35.186.224.45	TLSv1.2	82	Application Data
1925...	1300.593182	10.15.7.1	142.250.194.195	TCP	66	51952 → 443 [ACK] Seq=2601 Ack=12577 Win=131072 Len=0 SLE=16813 SRE=1...
1925...	1300.593277	10.15.7.1	142.250.194.195	TCP	66	[TCP Dup ACK 192511#1] 51952 → 443 [ACK] Seq=2601 Ack=12577 Win=13107...
1925...	1300.702471	10.15.7.1	142.250.194.195	TCP	54	51952 → 443 [ACK] Seq=2601 Ack=19637 Win=131072 Len=0
1925...	1300.703097	10.15.7.1	142.250.194.195	TCP	54	51952 → 443 [ACK] Seq=2601 Ack=22461 Win=131072 Len=0

Frame 192500: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface 0
Ethernet II, Src: AzureWaveTec_14:bb:47 (b4:8c:9d:14:bb:47), Dst: Cisco
Internet Protocol Version 4, Src: 10.15.7.1, Dst: 3.218.189.67
Transmission Control Protocol, Src Port: 51557, Dst Port: 443, Seq: 4939
Source Port: 51557
Destination Port: 443
[Stream index: 966]
[Conversation completeness: Complete, WITH_DATA (63)]
[TCP Segment Len: 0]
Sequence Number: 4939 (relative sequence number)
Sequence Number (raw): 2734488278
[Next Sequence Number: 4939 (relative sequence number)]
Acknowledgment Number: 124490 (relative ack number)
Acknowledgment number (raw): 16675168

Destination Port (tcp.dstport), 2 bytes

Packets: 311676 · Displayed: 67435 (21.6%) Profile: Default

5) What display filter would you use to show only DNS traffic?

Wireshark capture showing a filter `dns`. The packet list displays several DNS queries and responses. The packet details pane shows the structure of a DNS query for packet 3555.

No.	Time	Source	Destination	Protocol	Length	Info
3554...	2368.058122	1.10.10.10	10.15.7.1	DNS	152	Standard query response 0xf14a A applet-bundles.grammarly.net A ...
3554...	2368.058438	1.10.10.10	10.15.7.1	DNS	152	Standard query response 0x4839 A applet-bundles.grammarly.net A ...
3554...	2368.204196	1.10.10.10	10.15.7.1	DNS	173	Standard query response 0xfc8e HTTPS applet-bundles.grammarly.net A ...
3554...	2368.289398	10.15.7.1	172.16.1.80	DNS	78	Standard query 0xdc59 A capi.grammarly.com
3555...	2368.373427	172.16.1.80	10.15.7.1	DNS	267	Standard query response 0xa480 A roaming.officeapps.live.com CNA...
3555...	2368.373427	172.16.1.80	10.15.7.1	DNS	206	Standard query response 0xdc59 A capi.grammarly.com A 52.0.171.7...
3555...	2368.497461	10.15.7.1	1.10.10.10	DNS	94	Standard query 0x6566 A nav-edge.smartscreen.microsoft.com
3555...	2368.497688	10.15.7.1	1.10.10.10	DNS	94	Standard query 0x819a HTTPS nav-edge.smartscreen.microsoft.com
3555...	2368.557706	1.10.10.10	10.15.7.1	DNS	281	Standard query response 0x819a HTTPS nav-edge.smartscreen.micros...
3555...	2368.723645	1.10.10.10	10.15.7.1	DNS	217	Standard query response 0x6566 A nav-edge.smartscreen.microsoft...
3558...	2371.221085	10.15.7.1	172.16.1.80	DNS	76	Standard query 0x42d1 A in.grammarly.com
3558...	2371.224524	172.16.1.80	10.15.7.1	DNS	172	Standard query response 0x42d1 A in.grammarly.com A 3.210.94.245...
3558...	2371.615126	10.15.7.1	1.10.10.10	DNS	88	Standard query 0x973c A applet-bundles.grammarly.net

Ethernet II, Src: AzureWaveTec_14:bb:47 (b4:8c:9d:14:bb:47), Dst: Cisco
Internet Protocol Version 4, Src: 10.15.7.1, Dst: 1.10.10.10
User Datagram Protocol, Src Port: 63951, Dst Port: 53
Domain Name System (query)
Transaction ID: 0x6566
Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
Queries
nav-edge.smartscreen.microsoft.com: type A, class IN
[Response In: 355556]

Source Address (ip.src), 4 bytes

Packets: 356165 · Displayed: 4649 (1.3%) · Dropped: 0 (0.0%) Profile: Default

CN Assignment-2

- 6) How do you filter packets to display only those with a specific MAC address, such as `00:11:22:33:44:55`?

The image shows the Wireshark network traffic analysis interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains various icons for file operations, capture control, and analysis. The filter bar at the top shows the active filter: `eth.dst == 84-8C-9D-14-BB-47`.

The packet list pane displays a table of captured packets:

No.	Time	Source	Destination	Protocol	Length	Info
49	0.911902	142.250.195.35	10.15.7.1	TLSv1.2	1016	Application Data, Application Data
50	0.911902	34.128.9.3	10.15.7.1	TCP	68	443 → 50645 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1412 SACK...
51	0.911902	172.217.167.3	10.15.7.1	TCP	68	443 → 50601 [ACK] Seq=709 Ack=2 Win=290 Len=0 SLE=1 SRE=2
52	0.911902	142.250.77.202	10.15.7.1	TLSv1.2	338	Application Data
55	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	254	Application Data
56	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	293	Application Data
57	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	96	Application Data
58	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	131	Application Data
59	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	225	Application Data
60	0.912086	142.250.77.202	10.15.7.1	TLSv1.2	191	Application Data
135	1.409978	142.250.77.202	10.15.7.1	TCP	60	443 → 50445 [ACK] Seq=1148 Ack=36 Win=367 Len=0
136	1.409978	142.250.77.202	10.15.7.1	TCP	60	443 → 50445 [ACK] Seq=1148 Ack=75 Win=367 Len=0
137	1.409978	142.250.77.202	10.15.7.1	TCP	60	443 → 50445 [ACK] Seq=1148 Ack=110 Win=367 Len=0

The packet details pane shows the selected packet (No. 137) with the following fields:

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x80 (DSCP: CS4, ECN: Not-ECT)
- Total Length: 1002
- Identification: 0x7f28 (32552)
- > 0000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 116
- Protocol: TCP (6)
- Header Checksum: 0x6038 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 142.250.195.35
- Destination Address: 10.15.7.1
- Transmission Control Protocol Seq: 443, Dst Port: 50645, Seq: 1

The packet bytes pane shows the raw data of the selected packet, displayed in hexadecimal and ASCII.

At the bottom, the status bar indicates: Packets: 356165 · Displayed: 137987 (38.7%) · Dropped: 0 (0.0%) | Profile: Default

- 7) Write a display filter to show only packets where the destination IP address is `10.0.0.1`.

CN Assignment-2

Wireshark packet capture showing traffic filtered by `ip.dst == 10.0.0.1`. The packet list displays various ICMP and TCP packets. The packet details pane shows the structure of an Internet Protocol Version 4 packet.

No.	Time	Source	Destination	Protocol	Length	Info
404	3.270027	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
552	4.999869	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53110 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
553	5.000067	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53111 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
554	5.035836	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
555	5.035836	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
571	5.266201	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53114 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
575	5.295683	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
778	9.014780	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53111 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
779	9.014836	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53110 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
780	9.022270	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
781	9.029779	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)
798	9.280983	10.15.7.1	10.0.0.1	TCP	66	[TCP Retransmission] 53114 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=...
799	9.285239	172.16.0.1	10.15.7.1	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

Internet Protocol Version 4, Src: 10.15.7.1, Dst: 10.0.0.1

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 52
- Identification: 0x349a (13466)
- > 010. = Flags: 0x2, Don't fragment
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: TCP (6)
- Header Checksum: 0xab19 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.15.7.1
- Destination Address: 10.0.0.1

wireshark_Wi-Fi4XCCS2.pcapng | Packets: 830 · Displayed: 24 (2.9%) · Dropped: 0 (0.0%) | Profile: Default

8) How can you filter packets to display only those using the UDP protocol?

Wireshark packet capture showing traffic filtered by `udp`. The packet list displays various UDP packets. The packet details pane shows the structure of a User Datagram Protocol packet.

No.	Time	Source	Destination	Protocol	Length	Info
184	1.135593	fe80::9808:ff6f:d59...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
185	1.137044	10.15.7.60	224.0.0.251	MDNS	82	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
186	1.137364	fe80::9808:ff6f:d59...	ff02::fb	MDNS	102	Standard query 0x0000 PTR _googlecast._tcp.local, "QM" question
195	1.238339	10.15.7.66	224.0.0.251	MDNS	288	Standard query response 0x0000 PTR, cache flush Android-3.local ...
196	1.239433	fe80::dca1:8148:ca6...	ff02::fb	MDNS	308	Standard query response 0x0000 PTR, cache flush Android-3.local ...
198	1.326981	10.15.7.1	142.250.194.10	UDP	71	50452 → 443 Len=29
201	1.727930	10.15.7.1	142.250.194.10	UDP	71	50452 → 443 Len=29
217	2.386488	142.250.194.10	10.15.7.1	UDP	67	443 → 50452 Len=25
218	2.386488	142.250.194.10	10.15.7.1	UDP	68	443 → 50452 Len=26
219	2.386488	142.250.194.10	10.15.7.1	UDP	79	443 → 50452 Len=37
220	2.386488	142.250.194.10	10.15.7.1	UDP	77	443 → 50452 Len=35
221	2.386488	142.250.194.10	10.15.7.1	UDP	74	443 → 50452 Len=32
222	2.391691	10.15.7.1	142.250.194.10	UDP	313	50452 → 443 Len=271

> Frame 201: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on 0

> Ethernet II, Src: AzureWaveTec_14:bb:47 (b4:8c:9d:14:bb:47), Dst: Cisco_8

> Internet Protocol Version 4, Src: 10.15.7.1, Dst: 142.250.194.10

> User Datagram Protocol, Src Port: 50452, Dst Port: 443

- Source Port: 50452
- Destination Port: 443
- Length: 37
- Checksum: 0x1282 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 24]
- > [Timestamps]
- UDP payload (29 bytes)
- > Data (29 bytes)

User Datagram Protocol: Protocol | Packets: 830 · Displayed: 291 (35.1%) · Dropped: 0 (0.0%) | Profile: Default

CN Assignment-2

9) What filter would you use to show only ICMP (ping) packets?

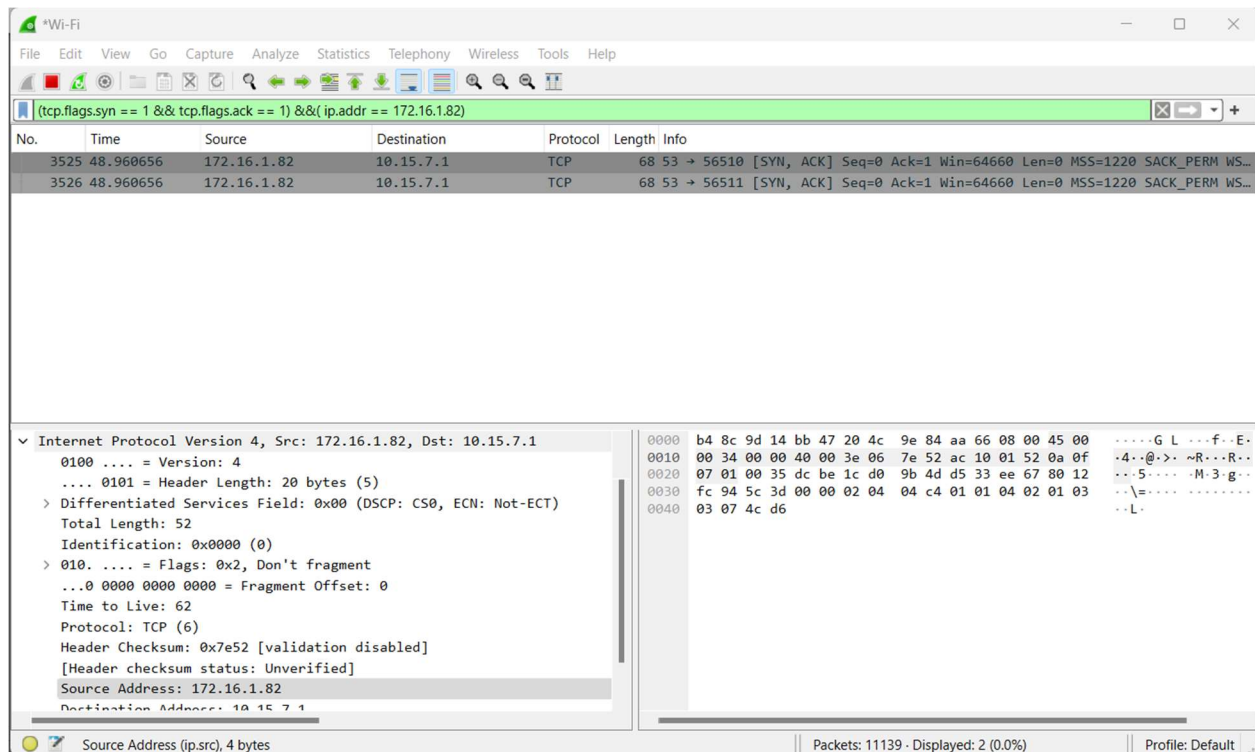
The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The filter bar at the top is set to "icmp". The packet list shows several ICMP Echo (ping) requests and replies between source 10.15.7.1 and destination 23.32.178.58. The selected packet (No. 2691) is an ICMP Echo (ping) reply. The packet details pane shows the following fields:

- Internet Protocol Version 4, Src: 10.15.7.1, Dst: 23.32.178.58
- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 60
- Identification: 0x03b3 (947)
- > 000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 128
- Protocol: ICMP (1)
- Header Checksum: 0x5ca4 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 10.15.7.1
- Destination Address: 23.32.178.58

The packet bytes pane shows the raw data in hexadecimal and ASCII. The status bar at the bottom indicates "Packets: 4007 · Displayed: 8 (0.2%)".

10) Write a display filter to show only packets that are part of an established TCP connection.

CN Assignment-2



- 11) How do you filter packets to show only those containing the word "login" in the HTTP request?
- 12) What filter would you use to show only packets from a specific subnet, such as `192.168.1.0/24`?
- 13) Write a display filter to show only packets that have been retransmitted.
- 14) How would you filter packets to show only those containing the ARP protocol?
- 15) What display filter would you use to show only SSL/TLS traffic?
- 16) List the different protocols that appear in the protocol column in the unfiltered packet-listing window in Wireshark GUI
- 17) How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received for the webpage you visited in your web browser? (By default, the value of the Time column in the packetlisting window is the amount of time, in seconds, since Wireshark tracing began. To display the Time field in time-of-day format, select the Wireshark View pull down menu, then select Time Display Format, then select Time-of-day.)
- 18) What is the Internet (IP) address of the URL you visited and what is the Internet address of your computer?
- 19) Print the two HTTP messages displayed in Wireshark GUI after you had visited above URL through your web browser. To do so, select Print from the Wireshark File command menu, and select "Selected Packet Only" and "Print as displayed" and then click OK.

CN Assignment-2

20) Capture three way handshake for the following link:

<https://docs.google.com/spreadsheets/d/1EO1pIEfwAMmKHXOdHMyTXEjlBm4kAUC8c9wqJ8i2NUk/edit?gid=0#gid=0>

No.	Time	Source	Destination	Protocol	Length	Info
2589	8.035764	172.16.1.82	10.15.7.1	TCP	60	[TCP Out-Of-Order] 53 → 54967 [FIN, ACK] Seq=129 Ack=46 Win=6476...
2590	8.035818	10.15.7.1	172.16.1.82	TCP	54	54967 → 53 [ACK] Seq=46 Ack=130 Win=131584 Len=0
2658	8.193467	172.16.1.82	10.15.7.1	DNS	114	Standard query response 0xb930 A sb.scorecardresearch.com A 0.0...
2663	8.195215	10.15.7.1	172.16.1.82	TCP	54	54966 → 53 [FIN, ACK] Seq=45 Ack=61 Win=131584 Len=0
2678	8.348321	172.16.1.82	10.15.7.1	TCP	68	[TCP Retransmission] 53 → 54971 [SYN, ACK] Seq=0 Ack=1 Win=64660...
2679	8.348362	172.16.1.82	10.15.7.1	TCP	60	53 → 54966 [FIN, ACK] Seq=61 Ack=46 Win=64768 Len=0
2680	8.348454	10.15.7.1	172.16.1.82	TCP	54	54966 → 53 [ACK] Seq=46 Ack=62 Win=131584 Len=0
2842	9.033037	10.15.7.1	172.16.1.82	TCP	66	54985 → 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2843	9.033170	10.15.7.1	172.16.1.82	TCP	66	54986 → 53 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
2879	9.166789	172.16.1.82	10.15.7.1	TCP	68	53 → 54985 [SYN, ACK] Seq=0 Ack=1 Win=64660 Len=0 MSS=1220 SACK_
2880	9.166789	172.16.1.82	10.15.7.1	TCP	68	53 → 54986 [SYN, ACK] Seq=0 Ack=1 Win=64660 Len=0 MSS=1220 SACK_
2890	9.166928	10.15.7.1	172.16.1.82	TCP	54	54985 → 53 [ACK] Seq=1 Ack=1 Win=131584 Len=0
2891	9.166961	10.15.7.1	172.16.1.82	TCP	54	54986 → 53 [ACK] Seq=1 Ack=1 Win=131584 Len=0